



HWA CHONG INSTITUTION
C1 Promotional Examination
Higher 2

NAME

CT GROUP

19S

CHEMISTRY

9729/03

Paper 3 Free Response

2 October 2019

55 min

Candidates answer on separate paper.

Additional Materials: Answer Paper, Cover Page and Data Booklet.

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Question 1 is a compulsory question.

Section B

Answer only **one** question from this section.

Begin each question on a **new piece** of paper. Leave **two blank lines in between your answers to each question part**.

The use of an approved scientific calculator is expected, where appropriate.

A Data Booklet is provided.

You are reminded of the need for good English and clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.

At the end of the examination, fasten all your work securely together, together with the cover page.

Section A

Question 1 is a **compulsory** question. All candidates should answer this question.

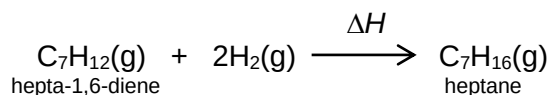
- 1 (a) The following three molecules have similar formulae but different shapes.



State their shapes and draw diagrams to illustrate your answers.

[3]

- (b) The hydrogenation of hepta-1,6-diene to heptane can be represented by the equation below.



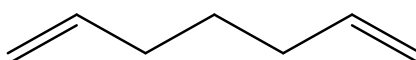
- (i) Explain in terms of the behaviour of the molecules, how an increase in pressure would increase the rate of this reaction. [1]
- (ii) Use the enthalpy changes of combustion below to calculate ΔH .

substance	$\Delta H_c / \text{kJ mol}^{-1}$
$\text{C}_7\text{H}_{12}(\text{g})$	-4467
$\text{H}_2(\text{g})$	-286
$\text{C}_7\text{H}_{16}(\text{g})$	-4781

[2]

- (iii) Although the hydrogenation of an alkene is a thermodynamically favorable reaction, it will **not** proceed without the addition of a catalyst. Suggest why this is so. [1]

- (c) The skeletal formula of hepta-1,6-diene is given below.



hepta-1,6-diene

- (i) Explain the origin of cis-trans isomerism in alkenes. [2]
- (ii) Compound **P** is an unbranched and non-cyclic constitutional isomer of hepta-1,6-diene. **P** has only two cis-trans isomers. Draw and label the cis and trans isomers of **P**. [2]
- (iii) Compound **Q** is another non-cyclic constitutional isomer of hepta-1,6-diene. **Q** exhibits cis-trans isomerism as well as enantiomerism. Suggest the structural formula for **Q**. [1]

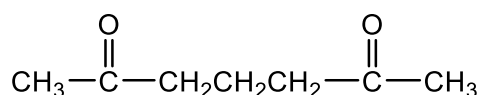
Alkenes can react with hot acidified potassium manganate(VII).

(iv) Write a balanced equation for the reaction between hepta-1,6-diene and hot acidified potassium manganate(VII), showing the structural formula of the organic product. Use [O] to represent the oxidising agent. [2]

(v) **R** and **S** are two other constitutional isomers of hepta-1,6-diene.

R has an unbranched and non-cyclic structure, and has four cis-trans isomers. 1 mole of **R** gives 2 moles of carbon dioxide when reacted with hot acidified potassium manganate(VII).

When **S** is reacted with hot acidified potassium manganate(VII), only one organic product, **T**, is obtained.

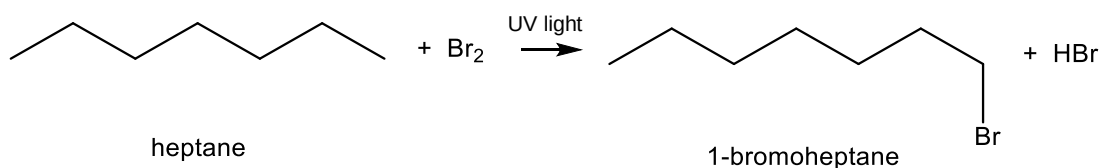


compound **T**

Suggest the structural formula for **R** and for **S**.

[2]

(d) Heptane and bromine can react to form 1-bromoheptane.



- Name the mechanism of this reaction.
- Use equations to describe the steps in the mechanism, naming each stage of the mechanism. You may represent heptane as RCH₃ and 1-bromoheptane as RCH₂Br.

[4]

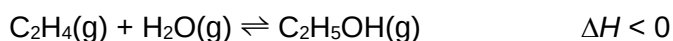
[Total: 20]

Section B

Answer **one** question from this section.

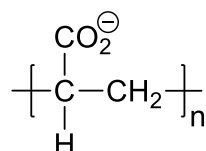
2 Ethanol, C₂H₅OH, is a volatile, flammable, colourless liquid with a slight characteristic odour. Owing to its antiseptic properties, ethanol is widely used in hand sanitisers.

- (a)** Industrially, ethanol is manufactured by passing a mixture of ethene and steam over a catalyst consisting of solid silicon dioxide coated with phosphoric(V) acid. The reaction is carried out at 300 °C and 60 atm.



- (i)** Draw the displayed formula of ethanol. [1]
- (ii)** State the bond angles of H–C–H and C–O–H in ethanol and account for any difference in their bond angles using the VSEPR theory. [3]
- (iii)** Use relevant bond energy values given in the *Data Booklet* to calculate the enthalpy change for this reaction, ΔH . [2]
- (iv)** Deduce how the value of the equilibrium constant would change, if at all, when the reaction is carried out at 150 °C and 60 atm instead. [2]
- (v)** State the reagents and conditions for the production of ethanol from ethene in the laboratory. [1]
- (b)** A typical hand sanitiser contains a polymer to give the hand sanitiser a gel-like texture and ethanol as its active ingredient.

The simplified structure of the anion in the polymer present in the hand sanitiser is given below.



Illustrate, with the aid of a suitable diagram, the type of bonding between the anion of the polymer and an ethanol molecule. [2]

- (c) In the manufacture of the hand sanitiser, the total volume of ethanol and water used must be greater than the capacity of the bottle as any combination of ethanol and water will result in a shrink in the total volume upon mixing.

(i) Suggest why there is a shrink in total volume upon mixing of ethanol and water. [1]

This shrink in the total volume can be determined from Fig. 2.1, which is the plot of the excess molar volume, V_m , of the mixture of ethanol and water against mole fraction of ethanol. The excess molar volume, V_m , can be given by the following equation:

$$\text{excess molar volume, } V_m = \frac{\text{change in total volume}}{\text{moles of ethanol} + \text{moles of water}}$$

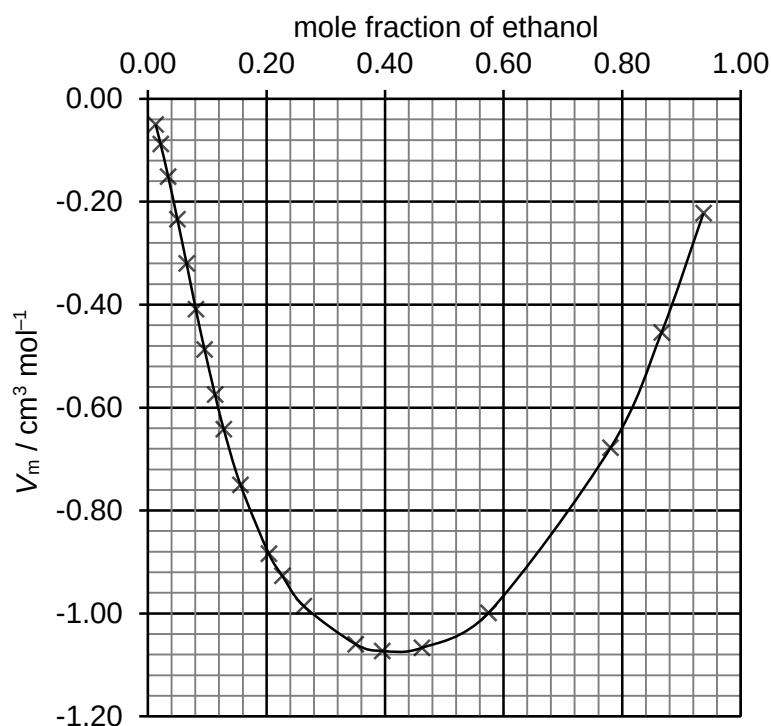


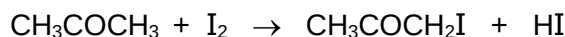
Fig. 2.1

In an attempt to produce a 200 cm^3 hand sanitiser, a manufacturer mixes 140 cm^3 of ethanol with 60 cm^3 of water.

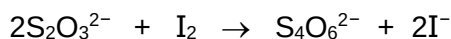
- (ii) Calculate the mole fraction of ethanol in the hand sanitiser. For the purpose of this calculation, you may assume that the hand sanitiser contains only ethanol and water. The densities of ethanol and water are 0.789 g cm^{-3} and 0.998 g cm^{-3} respectively. [2]
- (iii) Using Fig. 2.1 and your answer in (c)(ii), calculate the shrink in total volume when the ethanol and water are mixed. [1]

[Total: 15]

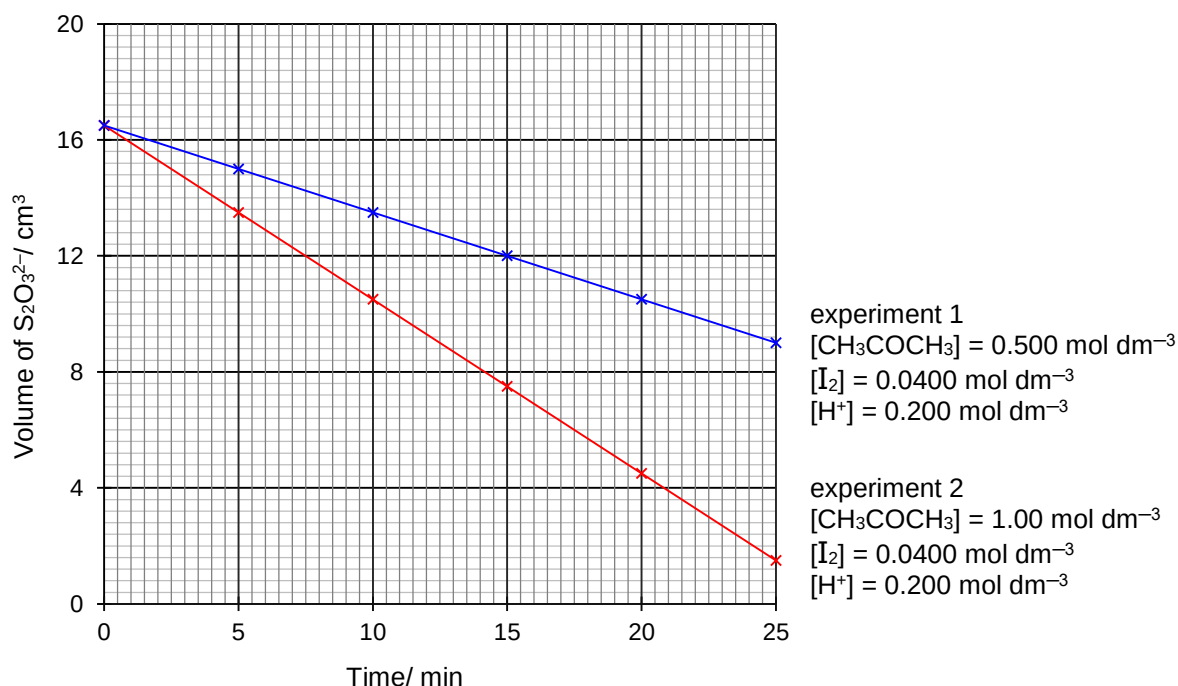
- 3 (a) Propanone and iodine react together in aqueous acidic solution according to the equation shown below:



To study the kinetics of this reaction, CH_3COCH_3 , I_2 and H^+ were mixed to give a 100 cm^3 reaction mixture. At fixed intervals of 5 min, 10.0 cm^3 of the reaction mixture was withdrawn and quenched. The mixture was then titrated quickly with $0.0500 \text{ mol dm}^{-3} \text{ Na}_2\text{S}_2\text{O}_3(\text{aq})$, using starch indicator.

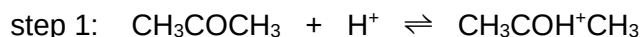


Two separate experiments were carried out with different concentrations of CH_3COCH_3 , while keeping the initial concentrations of H^+ and I_2 constant. The following results were obtained.

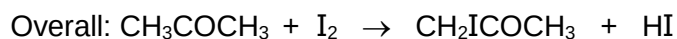
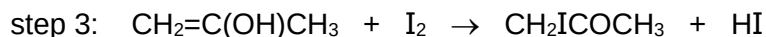


- Determine the order of reaction with respect to I_2 and to CH_3COCH_3 , showing your working clearly and explaining your answer. [3]
- The order of reaction with respect to H^+ was found to be one in another experiment. Using your answer to (a)(i), write the rate equation for the reaction. [1]
- From the graph, calculate the change in number of moles of I_2 per minute for experiment 1. [1]
- Using your answer in (a)(iii), find the initial rate in terms of $-\frac{d[\text{I}_2]}{dt}$ for experiment 1. [1]
- Hence, calculate the rate constant for experiment 1, stating its units. [2]

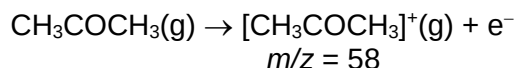
- (b) The proposed mechanism for the reaction between iodine and CH_3COCH_3 catalysed by acid is shown below:



step 2: (slow, rate-determining step)



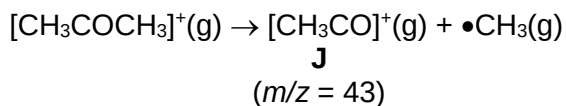
- (i) Write the balanced equation for the rate-determining step. [1]
- (ii) Hence, show whether the proposed mechanism is consistent with the rate equation obtained in (a)(ii). [1]
- (c) Propanone can undergo ionisation under suitable conditions as shown by the following equation, where m/z represents the mass-to-charge ratio of the species formed.



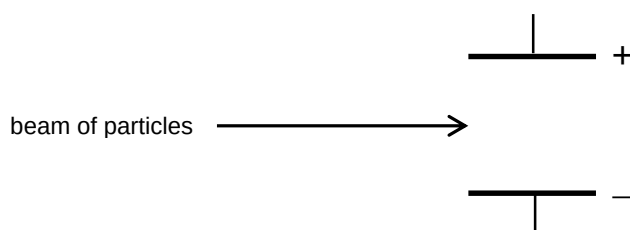
- (i) The above equation can be taken to represent the *first ionisation energy of propanone*.

Explain what is meant by the *first ionisation energy of propanone*. [1]

The ionised form of propanone may further break down to form **J** as shown below.



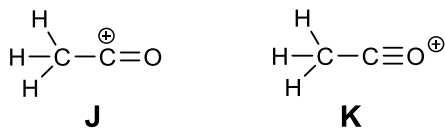
- (ii) A beam containing the ionised form of propanone and **J** is passed through an electric field as shown below.



Copy the diagram onto your writing paper and sketch on your diagram, the paths taken by the ionised form of propanone and **J**. Label the paths clearly. [2]

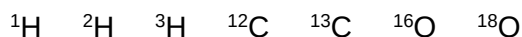
Question 3 continues on the next page.

- (iii) **J** has 6 electrons in the outermost shell of its positively charged carbon. **K** is another form of **J** where all the atoms are able to attain noble gas configuration.



Copy the structure of **J** onto your writing paper. By using a curly arrow to depict electron pair movement, suggest how **K** is obtained from **J**. Show any relevant lone pair in your diagram. [1]

- (iv) Some common isotopes of hydrogen, carbon and oxygen are shown below.



A species of $m/z = 59$ can also be detected when propanone is ionised. Suggest how this species might arise. [1]

[Total: 15]

~END OF PAPER~